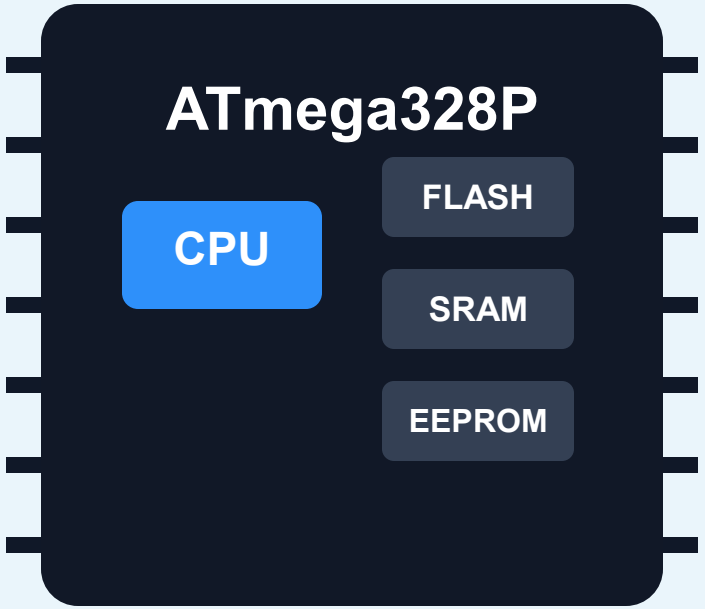
Inside a tiny computer

Introduction to microcontrollers and the 8-bit AVR



SESSION 1

Where is the computer?

A The whole board

connectors · regulator · USB interface · support parts

B The black chip

CPU · memory · peripherals · input/output

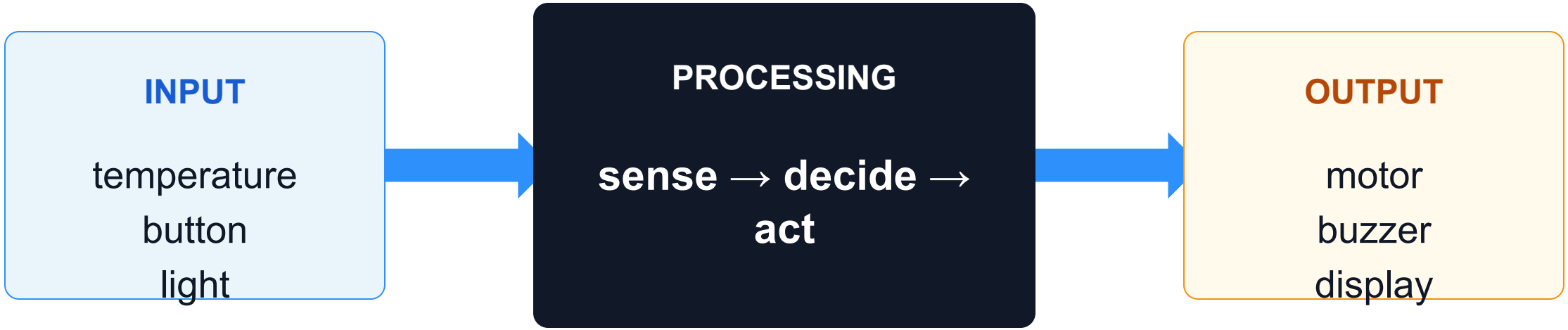


The chip is the microcontroller.
The board helps us power, program, and connect it.

Four things you should be able to do

01	**MODEL**	Trace input → processing → output and name the constraints.
02	**DISTINGUISH**	Separate a microcontroller, a microprocessor/SoC, and a development board.
03	**PREDICT**	Explain what Flash, SRAM, EEPROM, and reset do.
04	**BEGIN**	Screen a final-project idea before buying parts.

An embedded system closes a physical loop



CONSTRAINTS timing · memory · power · interfaces · cost

Classify by integration and system needs

MICROCONTROLLER

CPU + memory + peripherals on one chip

Dedicated control
Fast start
Tight power / cost

MPU / COMPLEX SoC

High compute; often external memory or a richer integrated system

Operating system
Large memory
Rich user interface

DEVELOPMENT BOARD

A device plus power, clock, programming, and connectors

Learning
Prototyping
Debug access

Boundaries blur. Choose the category that best explains the design trade-offs.

The Uno is the carrier; ATmega328P is the MCU

BOARD

power · USB · clock · reset · headers

CHIP

ATmega328P
CPU · memory · peripherals

OUR COURSE PATH

bare-metal AVR C · avr-gcc / avr-libc · direct register access

No Arduino API, Arduino framework, or C++ in submitted course code.

Quick prerequisite check

1 **$1011\ 0100_2$ = _____$_{16}$**

2 **In 0x20, which bit position is 1?**

3 Why does an LED need a series resistor?

4 Why must two connected circuits share ground?

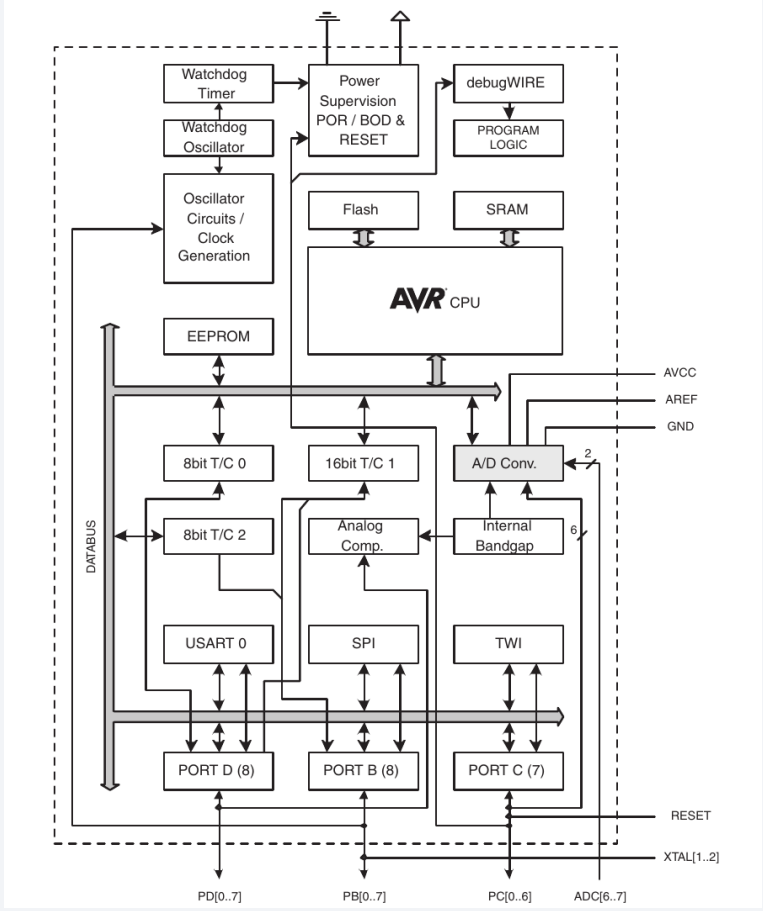
5 What does while (1) mean in C?

This is diagnostic—not graded.

ATmega328P is a complete control computer

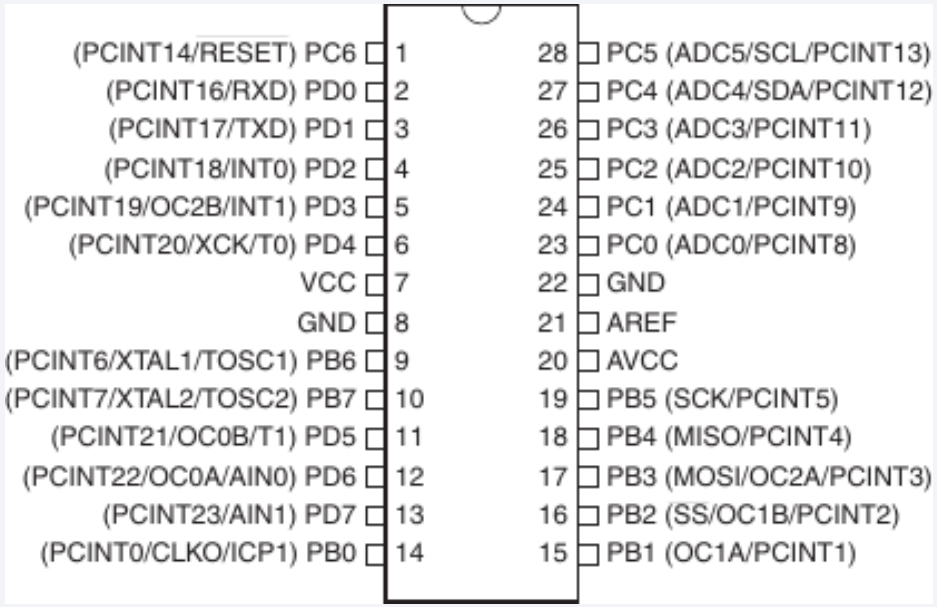
8-bit AVR CPU

32 KB Flash
2 KB SRAM
1 KB EEPROM
Timers · ADC · USART · SPI · TWI
Digital I/O and interrupt sources



These blocks communicate through internal buses.

A pin label is a menu of capabilities



PHYSICAL PIN 4

PD2

digital I/O

INT0 external interrupt

PCINT18 pin-change interrupt

Software configuration and circuit choices decide which role is active.

A bare chip needs five kinds of life support

1	**POWER + GROUND**	VCC and AVCC supplied; both GND pins connected
2	**DECOUPLING**	100 nF close to each supply pair; short current path
3	**CLOCK**	Internal oscillator or configured external clock source
4	**RESET**	Active-low reset with a defined high state
5	**PROGRAMMING**	ISP access: RESET, MOSI, MISO, SCK, VCC, GND

AREF is an analog-reference pin—not a supply pin to connect blindly to VCC.

Choose memory by what must survive

FLASH

PROGRAM

non-volatile

32 KB

instructions
constant tables

SRAM

WORKING STATE

volatile

2 KB

variables
stack

EEPROM

SAVED STATE

non-volatile

1 KB

calibration
settings

REGISTERS = the CPU’s fastest workbench for immediate operations and control

PREDICT BEFORE WE PRESS RESET

What will the chip forget—and remember?

LIVE · SRAM



D2 · b0 D3 · b1

SAVED · EEPROM



D4 · b0 D5 · b1

D13 MULTIMETER

5.00 V

Write three predictions before watching.

Live demo: make memory visible

TINKERCAD WIRING

D2–D3	SRAM counter LEDs
D4–D5	EEPROM counter LEDs
D13	multimeter red lead
GND	meter black + LED returns
RESET ↔ GND	momentary pushbutton

RUN THE EXPERIMENT

1 **Start simulation**

Watch the two counters and D13 voltage.

2 **Pulse RESET**

Do not stop/restart the simulator.

3 **Compare evidence**

Which state restarted? Which persisted?

The Uno is only the simulated ATmega328P carrier.

Three register writes become a voltage

```
DDRB  |= _BV(DDB5);

PORTB |=  _BV(PORTB5);
_delay_ms(500);

PORTB &= ~_BV(PORTB5);
```

direction

PB5 becomes an output

HIGH

D13 ≈ VCC

LOW

D13 ≈ 0 V

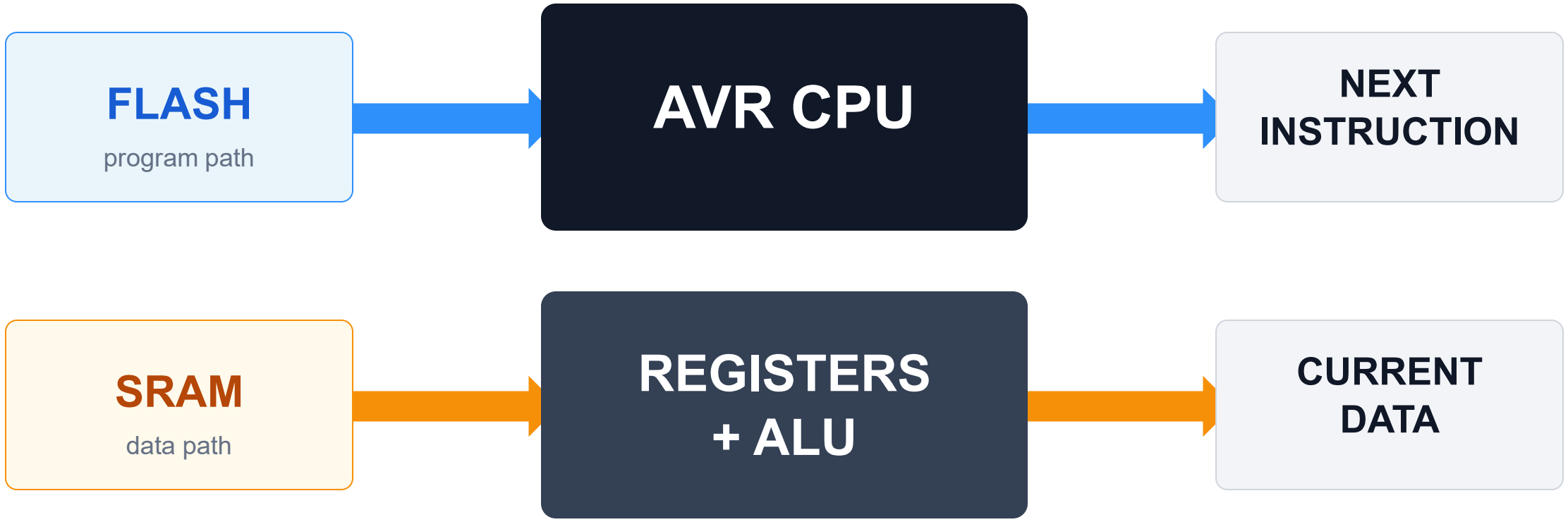
Preview only—you do not need to memorize the register syntax today.

The observations identify three different lifetimes

PROGRAM RESTARTS	**Flash retained the instructions**
LIVE COUNT → 00	**SRAM working state was initialized again**
BOOT COUNT + 1	**EEPROM retained saved state through reset**
D13: 0 V ↔ 5 V	**A register value became an electrical output**

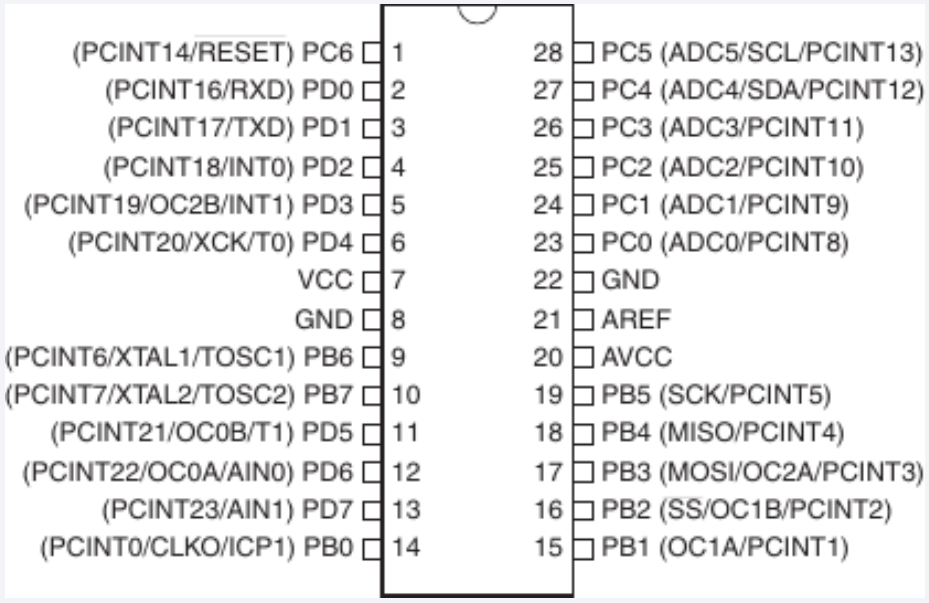
Evidence → explanation—not ‘the LEDs blinked, so it works.’

Separate paths keep instructions and data moving



Modified Harvard architecture: separate program and data spaces, with controlled ways to access both.

The datasheet is both a map and a contract



MAP

Which physical pin is PB5?

CAPABILITY

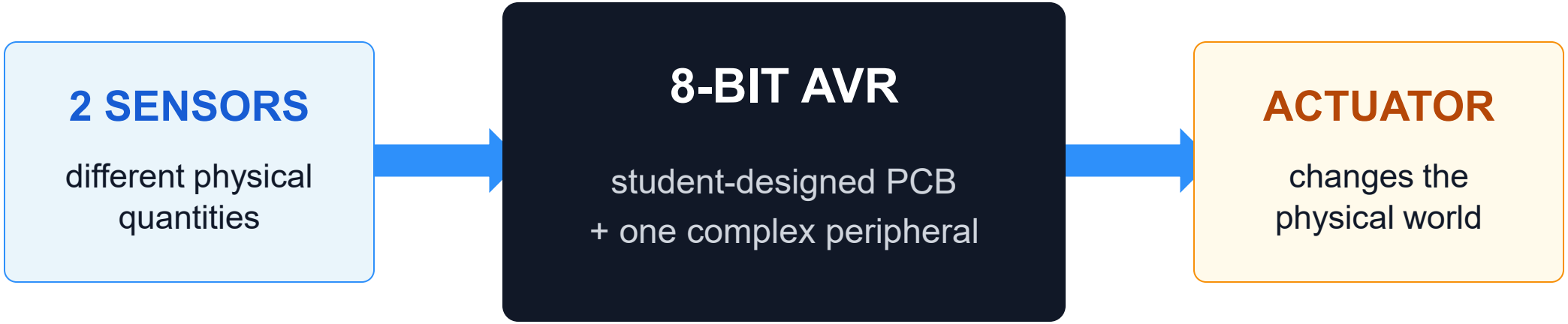
What alternate functions share PD2?

CONDITION

At what voltage and clock is operation specified?

ABSOLUTE MAXIMUM ≠ recommended operating target

Your final project is an embedded system



Complex peripheral: PWM, ADC, UART, I²C, or SPI

Teams of 2–3 · at least 30% original code · evidence and explanation required

WEEK 1 CHECKPOINT: two problem candidates + feasibility sketch

Six-minute design check: automatic plant watering

SCENARIO

A device measures soil moisture and room temperature, then runs a water pump when the plant needs water.

1. Label the inputs, processing rule, and output.
2. Place the program, live readings, and saved threshold in memory.
3. Name one timing, power, memory, or interface constraint.
4. **Which final-project requirements are met—and what is still missing?**

Work in pairs · one sketch · one spokesperson

Five ideas to carry into every later session

01 Embedded systems sense, decide, and act under constraints.

02 The ATmega328P is the MCU; a development board supports it.

03 Flash stores the program; SRAM works now; EEPROM remembers.

04 Power, ground, decoupling, clock, reset, and programming make a usable system.

05 **The datasheet—not a diagram copied from the internet—is the authority.**

NEXT: registers and the clock system